MSE 335 MATERIALS CHARACTERIZATION LABORATORY
SPR 2026, CRN: 19368

COURSE INSTRUCTOR INFORMATION:
Maria Okuniewski, Ph.D.
Purdue University

COURSE CREDIT HOURS: 3

COURSE MODALITY: In person

COURSE ATTENDANCE POLICY FOR LAB DAYS:
Laboratory attendance is mandatory for each lab. For students that are absent 25% per lab day will be deducted from your lab report grade. On-time attendance during your scheduled lab time is required. Attendance will be taken during your scheduled lab periods. If you are more than 5 minutes late for your scheduled time, you will lose 5% of your grade for each time you are late (up to 20% for the lab report). If you are 15 minutes later or more, this is counted as an absence (-25%).

However, you get one "free pass" for the entire semester. You can miss one lab with no deduction. This could be for sickness, conference, woke up late, etc… You will not be allowed to make up labs unless there are circumstances (excused absence regulations) which requires a letter from the Office of Dean of Students (ODOS). See the explanation below in ***Attendance Policy*.** It is the responsibility of the absent student to work with their lab partner(s) to determine what was missed in the lab and get the data for future data analysis.

An example of a lab grade calculation is given: In lab 1 the student missed one lab and was late for one lab; for lab 2 the student missed one lab; for lab 3 they were present and on time for all labs. This would result in a total maximum possible point value for lab report 1 of 100-5=95; for lab report 2 of 100-25=75; for lab report 3 – 100 possible points. (Note that in the third lab there are 5 lab days. All groups may not need the fifth lab and there is no penalty for not attending the last lab.)

In general, the course will follow the university guidance on attendance located at the end of this syllabus.

To accommodate lab and equipment capacities, as well as student progress, all lab activities will be on a scheduled basis. Weekly lab schedules/activities will be posted on Brightspace for each student. These schedules will be comprised of varying time increments for various labs in ARMS. Make sure to check this prior to your lab every week so that you know where to go.

If you cannot make your lab reservation/appointment due to an emergency, please notify your graduate TAs and Prof. Okuniewski via email as soon as possible.

COURSE ATTENDANCE POLICY FOR LECTURE DAYS:
Presence in the lecture is highly encouraged, but not required unless you are one of the scheduled

speakers for that day. If you are absent, it is your responsibility to ask classmates for missed material, not the instructors. If you are absent, you may miss important announcements, discussions, helpful hints for report writing, etc. Lectures will not be recorded. Cell phones shall be turned off during the lectures.

COURSE FORMAT:

The major graded items will be your laboratory reports from two main laboratory exercises and a final unknown project. Lab #1 will feature a metallic binary (copper-nickel) phase bolt. Lab #2 will feature an integrated circuit composed of non-metallic and metallic components that will be reverse engineered. Lab #3 will feature an "unknown" material intended to provide a unique sample preparation and characterization challenge. Each lab will have pre-lab assignments, including the submission of a draft report for only the first lab. Most of the lab activities will be conducted in teams of 2-3 and the associated data / outcomes may be shared with a larger study group or in some cases the entire class. The final unknown project will also be conducted in groups of 2-3.

In addition to lab reports, there is one major, non-lab project which culminates in an oral presentation to the class (safety). These presentations will occur during lectures on Friday. The safety presentations will be completed as a team of 2-3.

COURSE DESCRIPTION:

The principles of analytical methods for characterization of the structure and composition of materials will be covered. The specific techniques of interest include the following: optical microscopy, scanning electron microscopy, X-ray spectroscopy and diffraction, powder characterization, and stereology. Laboratory experiments consist of X-ray diffraction, optical microscopy, scanning electron microscopy, X-ray spectroscopy, metallographic preparation, and powder characterization. There will also be a student-designed laboratory for the identification of an unknown sample or characterization of an engineered product that will include a presentation of the results.

LEARNING RESOURCES

1) **Course Website**: BrightSpace

   → **Spring-2026-MSE-33500-001 LEC**

2) **Text book**: Microstructural Characterization of Materials, D. Brandon & W. Kaplan, Wiley 2008

   (Can download this for free from Purdue libraries.)

3) **Software**:

   a. Origin – Free for Purdue students – graphing software
   b. Other free software provided during the course for data analysis

4) **Use of Artificial Intelligence (AI) or Large Language Models (LLMs)**

The use of AI/LLM tools IS NOT PERMITTED IN THE COURSE FOR THE HOMEWORK AND THE FIRST TWO LAB REPORTS (Metals and Integrated Circuits). If AI/LLM use is detected, this can result in a zero for the assignment.

However, during the lab report for the Unknowns, the AI/LLM tools WILL BE ALLOWED.

For the Unknown lab report - AI tools should complement learning rather than substitute critical thinking or genuine engagement. Also, upholding academic integrity is crucial—utilize AI as a tool to augment understanding and creativity, avoiding plagiarism or misrepresentation. Additionally, exploring AI's impact within our field is encouraged for the Unknown lab. We will discuss the use of AI/LLMs during the introduction of the Unknown lab.

LEARNING OUTCOMES:

1. An understanding of the principles of basic techniques of optical microscopy, scanning electron microscopy, X-ray diffraction, spectroscopy, and stereology for the characterization of materials.
2. Understanding of the limitations of associated characterization techniques and ability to select the appropriate characterization technique based on material constraints.
3. Ability to write a technical report and give a technical presentation which includes analysis of new data and deriving conclusions supported by technical literature.
4. Ability to assess experimental methods to determine potential laboratory hazards and ability to propose alternatives or adjustments to minimize risk.

GRADE WEIGHTING:

Homework Assignments	24%
Lab Reports (3 x 22% / lab module)	66 %
Lab Safety Presentation	7 %
Participation / Data Sharing	3%

FINAL GRADING SCALE: A+ ≥98**,** A ≥93, A- ≥90, B+ ≥88, B ≥83, B- ≥80, C+ ≥78, C ≥73, C- ≥70, D+ ≥68; D ≥63; D- ≥60, F <60 or if caught in safety violation or cheating.

Assignments / due dates will be posted in Brightspace. Late submissions will be penalized by 10% per day, with penalty capped at 50%. Once the graded homework assignments have been returned, no further submissions will be accepted.

SCHEDULE:

Week	Dates	Okuniewski - Topic	Calendar Notes	Lab Schedule	Safety Presentations - Friday
1	1/12-1/16	Intro / Safety	No labs / Lectures only		
2	1/19-1/23	Sample Polishing / Etching	No Class Monday (MLK Day)	M (metal lab)	
3	1/26-1/30	Optical Microscopy / Stereology		M	
4	2/2-2/6	Scanning Electron Microscopy		M	1,2
5	2/9-2/13	X-Ray Diffraction		M	3,4
6	2/16-2/20	Lab Report Writing		C (circuit lab)	5,6
7	2/23-2/27	Spectroscopy / Lab Report Writing	M lab due	C	7,8
8	3/2-3/6	Lab Report Writing		C	9,10
9	3/9-3/13	Energy Dispersive Spectroscopy/ Statistics and Uncertainty		C	11,12
10	3/16-3/20	Spring Break!	Spring Break!		
11	3/23-3/27	Thermal Characterization / XRD Software		U (unknown lab)	13,14
12	3/30-4/3	Artifical Intelligence / Particle Size Analysis		U	15,16
13	4/6-4/10	Scattering (light, X-rays, neutrons)	C lab due	U	17,18
14	4/13-4/17	Advanced Characterization Tech		U	19
15	4/20-4/24	Final Project Discussion	Last week of lab access	U	
16	4/27-5/1	Student Presentations	Presentations during lab and lecture periods		
17	5/4-5/8	Exam Week	No final exam		

Monday lectures will usually be devoted to pre-lab in which all lab groups (Tuesday and Thursday) meet together to discuss background reading, lab procedures, characterization/prep techniques, and problems specific to the lab of that week.

Friday lectures will include data discussions from recently obtained data and discussions of characterization techniques, statistics, several communications workshops, and flash safety presentations given by students.

Tuesday and Thursday labs include three different lab experiment campaigns that will be carried out during the semester. Teaching assistants are responsible for conducting and advising students in the laboratory.

ATTENDANCE POLICY:

This course follows Purdue's academic regulations regarding attendance, which states that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor (Prof. Okuniewski) of the situation as far in advance as possible by email. (This will follow the "Course Attendance Policy for Lab Days" above.)

For unanticipated or emergency absences when advance notification to the instructor is not possible, the student should contact the instructor (Prof. Okuniewski) as soon as possible by email.

When the student is unable to make direct contact with the instructor and is unable to leave word with the instructor's department because of circumstances beyond the student's control, and in cases falling under excused absence regulations, the student or the student's representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted **by ODOS** for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent medical care. This communication should be as soon as possible to determine if and how you can make up the laboratory class.

For absences that do not fall under excused absence regulations (see above from ODOS), this course follows the following procedures:

1. Do not come to class if you are feeling ill. To accommodate these normal illness-related absences, you will be allowed to miss one lab (see **Course Attendance Policy for Lab Days).** Please email me at mokuniew@purdue.edu, with the subject line: MSE 335 absence. I do not need details about your symptoms. Just let me know you are feeling ill and cannot come to class. If it is an emergency situation, please follow the University regulations on emergent medical care (see above).
2. Unless it falls under the University excused absence regulations (see above), any work due should be submitted on time via our course Brightspace.
3. If that day's class involves assessed work such as a presentation, you and I will plan if and how you can make up the work, following the assignment guidelines. This plan must be done before the next class period, so again, email me immediately when you know that you will miss a class.
4. The most important consideration in any absence is how it will affect your achievement of the assignment objectives and the course learning outcomes.

ACADEMIC INTEGRITY:

Purdue University Regulations, Part 5, Section III-B-2-a describes the formal policies governing academic dishonesty. A guide providing specific examples, tips, and consequences is available at: https://www.purdue.edu/odos/osrr/resources/documents/academic_integrity.html. These rules cover all graded assignments and exams. Copying or sharing any part of the homework solutions from any source is a clear violation of university policy. Transcribing, paraphrasing, or following the outline of someone else's solution(s) is cheating. The test is simple: If you write down your solution after reading or hearing someone else's solution you are cheating. Likewise, if you share or exchange your solutions with another student, in written or spoken form, you are cheating.

Plagiarism and self-plagiarism are not permitted. If students are found to plagiarize or self-plagiarize material, they will be given a zero for the assignment. (Self-plagiarism is copying one's own work that could be in the form of a report, journal article, thesis, etc. and not providing appropriate citations.)

Students found to engage in academic dishonesty are subject to sanctions. For a first offense, the student will receive a zero score for the assignment, paper, report, quiz, or exam. Depending on the severity of the offense, the case may be referred to the Office of Student Rights and Responsibilities (OSRR) for

further action. A second offense will result in a failing grade for the course and definite referral to the OSRR. Students can also report issues of academic dishonesty that they observe, either through the OSRR by calling (765) 494-8778 or emailing integrity@purdue.edu . This information may be submitted anonymously. More details are available on our course Brightspace table of contents, under **University Policies.**

Use of AI/LLMs: The use of these tools should not take the place of your own writing, and you need to ensure that you also confirm facts that are generated, because these tools do not check the accuracy of their claims. You should acknowledge AI/LLM use (e.g. appropriately cite) where it has been used to assist in the process of creating your work. Inappropriate use of AI/LLM is viewed as plagiarism (see above paragraph). The use of AI/LLM tools that are permitted in this course are defined under the "Learning Resources" section. (May be used for only the Unknown Lab Report). If instructors suspect that you are trying to present or submit AI/LLM output as your own work (we do have AI/LLM checkers), then the regulations described above will be applied.

PURDUE HONOR'S PLEDGE:
As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together – we are Purdue.

NONDISCRIMINATION STATEMENT:
Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. More details are available on our course Brightspace table of contents, under University Policies.

ACCESSIBILITY:
Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at:**drc@purdue.edu** or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must send me your Course Accommodation Letter. Instructions on sharing your Course Accommodation Letter can be found by visiting: **https://www.purdue.edu/drc/students/course-accommodation-letter.php** Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodations

MENTAL HEALTH/WELLNESS STATEMENT:
If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try try Therapy Assistance Online (TAO), a new web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to students, faculty, and staff at any time.
If you need support and information about options and resources, please contact or see the Office of

the Dean of Students (ODOS). Call 765-494-1747.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the session. Sign up is free and can be done on BoilerConnect.
If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays.

The Community, Assistance and Resources for Engineering Students (CARES) Hub is available to support the well-being of all engineering students. The CARES Hub, located in ARMS 1261 and 1264, is a welcoming, inclusive space for students to study, connect, grab a snack, and relax during the week. CARES also offers wellness activities, educational workshops, peer mentoring, social events, and on-site counseling. If you need help handling stress or working through a problem, schedule time with our CARES Hub therapist, Jennie Beutler, here or drop by ARMS 1251 to speak to Jennie without an appointment during her **Walk-In Hours** from 1:00 – 2:00 p.m. Monday-Friday. Please visit the CARES Hub webpage to learn more. The CARES staff look forward to connecting with you!

BASIC NEEDS SECURITY:
Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday. Considering the significant disruptions caused by the current global crisis as it relates to COVID-19, students may submit requests for emergency assistance from the Critical Need Fund. The ACE Campus Food Pantry is open to the entire Purdue community. The Center for Advocacy, Response & Education (CARE) (open to all Purdue students) provides support and advocacy for survivors of sexual violence, dating violence, and stalking.

EMERGENCY PREPARATION:
In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

SAFETY:
Safe procedures are to be followed at all times. Failure to comply will result in expulsion from the laboratory rooms until proper adjustments are made. Work missed during this time is the student's responsibility alone and might not be recoverable. Multiple safety violations may, at the discretion of the instructor(s), result in grade reductions (possibly to zero) and/or longer expulsion from laboratory environments (possibly for the entire course).

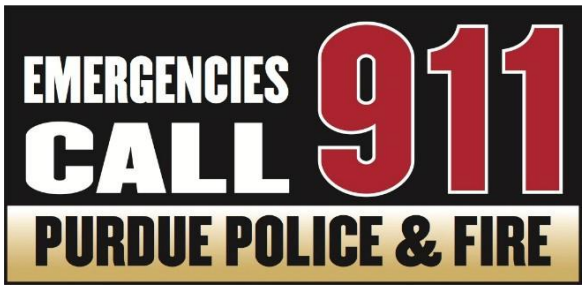
Report any breakages, unsafe process observations, etc., to an instructor as soon as practical. Failure to report these is considered a major safety violation.

EMERGENCY PREPAREDNESS:
Emergency preparedness is your personal responsibility. Purdue University is actively preparing for natural disasters or human-caused incidents with the ultimate goal of maintaining a safe and secure campus. For full details, visit: http://www.purdue.edu/ehps/emergency_preparedness/

- For any emergency call 911.
- There are nearly 300 Emergency Telephone Systems throughout campus that connect directly tothe Purdue Police Department (PUPD). If you feel threatened or need help, push the button and you will be connected to the PUPD.
- If we hear a fire alarm we will immediately evacuate the building and proceed outside. Do notuse the elevator.
- If we are notified of a Shelter in Place requirement for a tornado warning we will shelter in the lowest level of this building away from windows and doors.
- If we are notified of a Shelter in Place requirement for a hazardous materials release or civildisturbance, we will shelter in our classroom shutting any open doors and windows.

ARMS Building Emergency Plan
https://www.purdue.edu/ehps/emergency_preparedness/bep/arms-bep.html

HAMP Building Emergency Plan
Purdue University - Hampton Hall Building Emergency Plan (HAMP)



EMERGENCY PREPAREDNESS SYLLABUS ATTACHMENT

EMERGENCY NOTIFICATION PROCEDURES are based on a simple concept – if you hear a fire alarm inside, proceed outside. If you hear a siren outside, proceed inside.

- **Indoor Fire Alarms** mean to stop class or research and immediately **evacuate** the building.
  - Proceed to your Emergency Assembly Area away from building doors. **Remain outside** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.
- **All Hazards Outdoor Emergency Warning Sirens** mean to immediately seek shelter (**Shelter in Place**) in a safe location within the closest building.
  - "Shelter in place" means seeking immediate shelter inside a building orUniversity residence.

    This course of action may need to be taken during a tornado, an active threat including a shooting or a release of hazardous materials in the outside air. Once safely inside, find out more details about the emergency*. **Remain in place** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

**In both cases, you should seek additional clarifying information by all means possible…Purdue Emergency Status page, text message, Twitter, Desktop Alert, Albertus Beacon, digital signs, email alert, TV, radio, etc.…review the Purdue Emergency Warning Notification System multi-communication layers at http://www.purdue.edu/ehps/emergency_preparedness/warning-system.html*

EMERGENCY RESPONSE PROCEDURES:

- Review the **Emergency Procedures Guidelines** https://www.purdue.edu/emergency_preparedness/flipchart/index.html
- Review the **Building Emergency Plan** (available on the Emergency Preparedness website or from the building deputy) for:
  - evacuation routes, exit points, and emergency assembly area
  - when and how to evacuate the building.
  - shelter in place procedures and locations
  - additional building specific procedures and requirements.

EMERGENCY PREPAREDNESS AWARENESS VIDEOS

- **"Run. Hide. Fight.®"** is a 6-minute active shooter awareness video that illustrateswhat to look for and how to prepare and react to this type of incident. See: https://www.youtube.com/watch?v=5mzI_5aj4Vs (Link is also located on the EP website)